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GRU-Based Interpretable Multivariate Time Series Anomaly Detection in Industrial Control System

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ABSTRACT

Interpretable multivariate time series anomaly detection is an important technology to prevent accidents and ensure the reliable operation of Industrial Control Systems. A key limitation lies in the lack of a model to achieve better detection performance and more reliable interpretability, and keep a balance between performance efficiency and training optimization. In this paper, we propose GRN, an Interpretable Multivariate Time Series Anomaly Detection method based on neural graph networks and gated recurrent units (GRU). GRN can automatically learn potential correlations between sensors from multidimensional industrial control time series data, quickly mine long-term and short-term dependencies, to improve detection performance and help users to infer the root cause of detected anomalies. Based on GRU, GRN preserves the original advantages of processing the sequences and capturing the time series dependencies, moreover solves the problem of gradient disappearance and gradient explosion. We compare the performance of nine state-of-the-art algorithms on two real water treatment datasets (SWaT, WADI). GRN achieves better detection precision and recall. Meanwhile, the comparison of Area Under the Curve (AUC) demonstrates that GRN has the effect of maintaining balance between detection performance and training optimization. Compared with a Graph Deviation Network(GDN), GRN has achieved greater interpretability.

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1. Introduction

Interpretable multivariate time series (MTS) anomaly detection is an important technology to prevent accidents and ensure the reliable operation of Industrial Control Systems (ICS) (Lyu et al., 2019; Yu et al., 2021). Anomaly detection is the process of identifying abnormal events or behaviors from normal time series. We devide multivariate time series anomaly detection methods into statistics-based, machine learning-based, and deep learning-based. The statistical-based anomaly detection method learns a generative model that fits a given dataset and then identifies objects in the low-probability region of the model as outliers (Wang et al., 2011; Xu et al., 2020). Machine learning-based methods use algorithms, such as isolation forest (iForest) (Liu et al., 2012), Linear Regression, clustering (Xu et al., 2021a,b), Support Vector Machine(SVM) and others, to classify or identify anomalies as much as possible. Deep learning-based methods use deep neural network to predict multidimensional time series or reconstruct normal time series model, including Long Short-Term Memory (LSTM) (Niu et al., 2020), GAN (Li et al., 2019), autocoder (Sakurada and Yairi, 2014) and so on, to realize anomaly detection. Nowadays, interpretability is necessary for locating anomalies and helping users identify affected sensors in multivariate time series anomaly detection, such as, Deng and Hooi (Deng and Hooi, 2021) proposed a Graph Deviation Network (GDN) to learn dependencies between variables, identify and interprete abnormal behaviors automatically.

The above research has achieved periodic results, such as improving attack detection precision, partially achieving interpretability, and so on. However, there are still some deficiencies to be solved.

- The anomaly detection ability needs to be improved. The current anomaly detection is not an entirely satisfactory result for the ICS. The anomaly detection methods based on statistics and machine learning are susceptible to noise in anomaly

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detection, and the detection effect is unstable. Most deep learning-based anomaly detection methods are robust against noise, and such methods can model complex patterns in various domains by extending to higher dimensions. However, they have insufficient ability to mine potential correlations deeply from multivariate time series, leading to a lower detection precision and recall.

- Insufficient ability for interpretability in industrial control system. In the field of industrial control system security, it is not only necessary to achieve better detection results, but also to be able to explain which components are attacked, for intercepting the attack in time. Most anomaly detection schemes for industrial control systems only consider the detection effect, ignoring the interpretation ability. GDN has some interpretation capability on the SWaT and WADI datasets, while its interpret ability need to be improved.
- The need for balance between performance efficiency and training optimization. Although increasing the number of layers in the neural network can improve the model performance, the difficulty of training and optimization also increases, meanwhile the problems of gradient disappearance and gradient explosion begin appear. Both problems are affected by the cumulative multiplication of gradients in back-propagation, leading to shallow network parameters, including small changes or large shocks.

Inspired by the above problem, we propose a GRN, GRU-Based Interpretable Multivariate Time Series Anomaly Detection in Industrial Control System, which incorporates neural graph networks based on attention mechanism for automatically learning sensors relation graphs, and GRU (Cho et al., 2014) for identifying and interpreting abnormal behavioral features. GRN has the following advantages.

- Effective. Compared to multidimensional time series data anomaly detection algorithms in industrial control systems, such as MERLIN (Nakamura et al., 2020), LSTMNDT (Hundman et al., 2018), DAGMM (Zong et al., 2018), OmniAnomaly (Su et al., 2019), MAD-GAN (Li et al., 2019), USAD (Audibert et al., 2020), MTAD-GAT (Zhao et al., 2020), CAE-M (Zhang et al., 2021), GDN (Deng and Hooi, 2021), GRN achieves higher precision and recall.
- Better interpretability ability. In terms of interpretability, it can effectively acquire complex dependencies between sensors to assist users in judging and locating anomalies.
- Ability to solve the problem of gradient disappearance and gradient explosion. Based on GRU algorithm, GRN preserves the original advantages of processing the sequences and capturing the time series dependencies, and solves the problem of gradient disappearance and gradient explosion. Compared to GDN, GRN has the effect of maintaining balance between detection performance and training optimization.

The remainder of this paper is organized as follows. Section II provides an overview of related work. Section III provides our approach in details. Section IV explores performance evaluation and anomaly interpretability of the proposed approach. Conclusions and suggestions for future work are presented in Section VI.

2. Related work

2.1. Statistics-based anomaly detection

The statistical-based anomaly detection methods learn a generative model that fits a given dataset and then identifies objects in the low-probability region of the model as outliers. Jones et al. (Jones et al., 2016) extract Statistically Smoothed Trajectory (SST) features of time series and utilized a set of nonlinear functions to model the relevant dimensions to detect anomalies. Wang et al. (Wang et al., 2011) propose a statistical anomaly detection method based on Tukey and relative entropy statistics to analyze and make decisions on monitoring data from past periods. Kindl et al. (Kindl et al., 2018) propose a Low-high pass filter that detects outliers using historical values of the mean and standard deviation. Wun et al. (Huo et al., 2019) propose a statistical distribution outlier detection method based on Euclidean distance, which identifies data with large deviations from the center as outliers. Nakamura et al. (Nakamura et al., 2020) propose an anomaly detection algorithm, MERLIN, to find discordances of various lengths in a time series by sliding along the sequence window and combining the distances to examine each subsequence.

2.2. Machine learning-based anomaly detection

Anomaly detection methods based on machine learning use methods such as model, distance, or density to classify and identify anomalies as much as possible. For example, Liu et al. (Liu et al., 2012) propose a sequence data anomaly detection method based on isolation forest (iForest). According to selecting features randomly, a binary tree was builded for the samples. The data farther from the root is normal data, and the closer was identified to abnormal. Angiulli and Pizzuti (Angiulli and Pizzuti, 2005) propose an unsupervised distance-based outlier detection method, and use the sum of the distances of the k nearest neighbors of a point to detect outliers. Zeng et al. (Zeng et al., 2021) propose a HTM-based method for multivariate time series anomaly detection, which deals with data in multivariate time series by using multiple spatial pools instead of the traditional structure with a single spatial pool and encoder. Xu et al. (Xu et al., 2021a,b) construct a behavioral model based on secondary labeling of states-duration evolution graph and state transition probability graph to detect semantic attacks.

2.3. Deep learning-based anomaly detection

Deep learning-based methods use deep neural network to predict multidimensional time series or reconstruct normal time series model. VAE-GAN (Niu et al., 2020) based on Long Short Term Memory(LSTM) exploits both the mapping ability of the encoder and the discriminative ability of the discriminator, in which, LSTM acts as the encoder, generator, and discriminator. This model uses the reconstruction error and the discrimination result to optimize and improve the detection precision. MAD-GAN (Li et al., 2019) adopts the LSTM network as the base model to capture the temporal correlation of time series data. In addition, considering the complex dependencies between different time-series dimensions, generative adversarial networks (GANs) is used for unsupervised anomaly detection. Aggarwal (Aggarwal, 2017) uses the reconstruction error based on an autoencoder(AE) to measure anomaly. DAGMM (Zong et al., 2018) consists of a compression network and an estimation network that combines a deep autoencoder (DAE) density estimation process for end-to-end joint training. TCN (Bai et al., 2018) treats high-dimensional data as tensor inputs and considers large receptive fields through dilated convolutions. LSTNet (Lai et al., 2018) adoptes Convolutional Neural Networks(CNN) and Recurrent Neural Networks(RNN) to extract short-term local dependency patterns among variables, for discovering long-term ways of time series. According to modeling the data distribution through random latent variables, OmniAnomaly (Su et al., 2019) constructs a stochastic recurrent neural network to acquire the normal patterns of multivariate time series. DOMINANT (Ding et al., 2019) trains a graph convolutional network

with a deep autoencoder to achieve anomaly detection with measuring the reconstruction error of the structure and attributes of the graph. DeepState (Rangapuram et al., 2018) combines a state-space model with a deep recurrent neural network to learn the parameters of the entire network by maximum log-likelihood. DeepGLO (Sen et al., 2019) utilizes both global and local features during training and prediction. Its global components are based on matrix factorization, which can acquire global patterns by describing each time series as a linear combination of fundamental elements. USAD (Audibert et al., 2020) is based on autoencoders and also uses an adversarial training strategy to amplify the reconstruction error of anomalous inputs. MTAD-GAT (Zhao et al., 2020) is a prediction-based model based on a graph attention network that models the correlation between and within each time series. CAE-M (Zhang et al., 2021) describes complex spatio-temporal patterns by performing simultaneous reconstruction and predictive analysis, and it also introduces an autoregressive linear model. GDN (Deng and Hooi, 2021) is a graphical deviation network for learning dependencies between variables and automatically identifying abnormal behavior.

The current anomaly detection is not an entirely satisfactory result for the ICS. The anomaly detection methods based on statistics and machine learning are susceptible to noise in anomaly detection, and the detection effect is unstable. Most deep learning-based anomaly detection methods are robust against noise, and such methods can model complex patterns in various domains by extending to higher dimensions. However, they have insufficient ability to mine potential correlations deeply from multivariate time series, leading to a lower detection precision and recall. Moreover, most of above methods have no ability to help users find the actual root cause of anomalies, and locate anomalous sensors. Although GDN has some capability to locate anomalous sensors, its interpret ability need to be improved. In this paper, we propose GRN, a interpretable multivariate time series anomaly detection method based on neural graph networks and GRU. GRN can automatically learn potential correlations between sensors from multi-dimensional industrial control time series data, quickly mine long-term and short-term dependencies, to improve detection performance and help users to infer the root cause of detected anomalies. Based on GRU, GRN preserves the original advantages of processing the sequences and capturing the time series dependencies, moreover solves the problem of gradient disappearance and gradient explosion. Compared with nine state-of-the-art algorithms on two real water treatment datasets (SWaT, WADI), GRN achieves better detection precision and recall. Meanwhile, the comparison of Under the Curve (AUC) demonstrates that GRN has the effect of maintaining balance between detection performance and training optimization. GRN has achieved greater interpretability than the traditional interpretable algorithm, such as GDN.

3. Approach

This section details the proposed GRN model for multivariate time series anomaly detection. Figure 1 provides the overall network architecture of the proposed GRN. Specifically, GRN consists of four modules: (1) Learning Sensor Embedding. (2) Graph Structure Learning. (3) Feature Behavior Prediction. (4) Abnormal Score. The details of each module are as follows:

3.1. Learning Sensor Embedding

We prefer to represent each sensor in a more flexible way to capture the behavioral characteristics of the sensor in a multivariate way. We denote the set of these sensor embeddings as $V = \{v_1, \ldots, v_N\}$. The behavioral characteristics of each of the N sensors are captured by introducing and embedding vector v_i for each sensor to represent its features. The behavioral properties of these sensors will be correlated in a particular way. Then, after these embedding vectors are randomly initialized, they are trained with the rest of the model. Where $v_i \in R^d, \forall i \in \{1, \ldots, N\}$ corresponds to the dimension of the embedding vector.

The similarity between these v_i indicates the similarity in behavior, and these embedding vectors will use for the following two modules:

(1) Graph-oriented structure learning to determine which sensors are related to each other.

(2) In the attention mechanism, attention is paied to adjacent sensors in a way that allows different types of sensors to produce heterogeneous effects.

3.2. Graph Structure Learning

The main goal of this paper is to learn the graph structure-oriented, using the embedding vectors of the sensors to understand the dependencies between the sensors. We consider that the dependency patterns between different embedding vectors are different. Therefore, we transform it into a directed graph structure G (adjacency graph). The method is achieved by learning a sparse critical matrix A using the top k neighbors of each sensor i and computing the similarity of a node's embedding vector to its neighbors. Among them, nodes represent sensors, and edges represent dependencies between them. The adjacency matrix is denoted by A. A_{ij} is denoted by the existence of a directed edge from node i to node j, and the parameter k controls the sparsity of the generated adjacency graph G.

$$e_{ji} = \frac{v_i^T v_j}{\|v_i\| \cdot \|v_j\|} \tag{1}$$

where e_{ij} is the normalized dot product between the embedding vector of sensor i and the candidate relation $j \in C_i$, which is the cosine similarity between any two sensors i and j.

3.3. Feature Behavior Prediction

To get which sensors behave abnormally and how they deviate from normal behavior, we need interpretability for the anomalies detected in anomaly detection. In this module, we employ a prediction-based approach. The method predicts the expected behavior of each sensor at each time based on history, which helps to subsequently identify sensors with considerable deviations from their expected characteristic behavior. By comparing the expected behavior of each sensor with the observed behavior, the user can understand the gradual transformation of the sensor into an anomaly.

Given a fixed window size $w > 0$ of all historical sensor time-series data at time steps $(t - w)$ to $(t - 1)$, define the model input $x_i^{(t)} = \left[s_i^{(t-w)}, s_i^{(t-w+1)}, \ldots, s_i^{(t-1)}\right]$, $x_i^{(t)} \in R^w$.Our model needs to predict the target output's current moment sensor data s_i^t.

Feature extractor. Our model captures the relationship between sensors by introducing a graph attention-based feature extractor, adding sensor embedding vectors v_i. That represents different behavioral features of different types of sensors and fusing nodes with their adjacencies according to the learned graph structure Feature information of the node. The aggregate representation formula is as follows:

$$z_i^{(t)} = ReLU\left(\alpha_{i,i} W x_i^{(t)} + \sum_{j \in \mathcal{N}} \alpha_{i,j} W x_j^{(t)}\right) \tag{2}$$

where $\mathcal{N}(i) = \left\{j \middle| A_{ji} > 0\right\}$ is denoted as the adjacency set of node i obtained from the learned adjacency matrix A, $W \epsilon R^{d \times w}$ is the

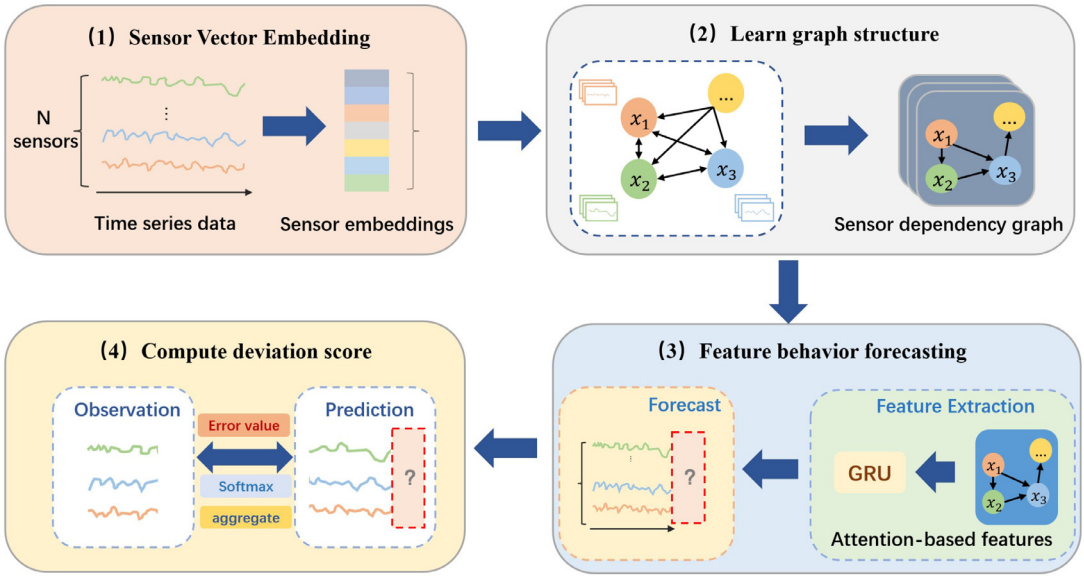


Fig. 1. The overall framework of GRN.

trainable weight matrix. It uses for a shared linear transformation applied to each node. The calculation of the attention coefficient $\alpha_{i,j}$ is as follows:

$$g_i^{(t)} = concat\left(v_i, Wx_i^{(t)}\right) \tag{3}$$

$$\pi(i,j) = LeakyReLU\left(a^T\left(g_i^{(t)} \bigoplus g_j^{(t)}\right)\right), j \in \mathcal{N}(i) \tag{4}$$

$$\alpha_{i,j} = \frac{exp(\pi(i,j))}{\sum_{k\epsilon\mathcal{N}(i)\cup\{i\}} exp(\pi(i,k))} \tag{5}$$

where $concat(\cdot)$ means concatenation, $g_i^{(t)}$ means concatenating the sensor embedding v_i with its corresponding transformation feature $Wx_i^{(t)}$, and $\alpha_{i,j}$ is the learning coefficient vector in the attention mechanism. *LeakyReLU* is used as a nonlinear activation function to calculate the attention coefficient. The problem of gradient disappearance is mitigated by this weight-initialized activation function. Compared to the *ReLU* function, *LeakyReLU* solves the neuron death problem of *ReLU* while inheriting the advantages of the *ReLu* function. After calculating the attention coefficients, we use the softmax function to normalize the attention coefficients.

Gated Recurrent Unit

We take feature nodes $z^{(t)} = [z_1^{(t)}, z_2^{(t)}, ..., z_N^{(t)}]$ as input to a gated recursive unit (GRU), which contains reset gates and update gates can easily preserve existing specific features and capture the dependencies present in time series features in depth. GRU controls the gradient by controlling the update gate in the structure to solve the gradient disappearance and gradient explosion problems.

The internal structure of the GRU is shown in Figure 2, which contains two gates, the reset gate, and the update gate. There are two gates which are the reset gate and the update gate. The reset gate is responsible for forgetting the amount of information in the past. The smaller the reset gate, the less information about the previous state is written, and the update gate is similar to the forget gate and input gate of LSTM, which are used to control the previous form. Its function update is used to control the degree to which the state information of the previous moment is brought into the current state. This way determines what information is to be forgotten and new information is added. The larger the value of the update gate, the more the state information of the previous moment is brought in. Reset gates and update gates help capture long-term and short-term dependencies, respectively. The GRU unit calculates $z^{'(t)}$ by the following formula:

$$r_t = \sigma\left(z_t W_{zr} + H_{t-1} W_{hr} + b_r\right) \tag{6}$$

$$\rho_t = \sigma\left(z_t W_{z\rho} + H_{t-1} W_{h\rho} + b_\rho\right) \tag{7}$$

$$\widetilde{H}_t = \tanh\left(z_t W_{hz} + R_t \odot H_{t-1} W_{hh} + b_h\right) \tag{8}$$

$$H_t = (1 - P_t) \odot H_{t-1} + P_t \odot \widetilde{H}_t \tag{9}$$

$$z^{'(t)} = H_t \tag{10}$$

Among them, r_t and ρ_t are the gate values of the reset gate and update gate, respectively, W represents the weight matrix, and the two gate values are calculated by linear transformation using H_{t-1} and z_t splicing to obtain them. R_t and P_t are their sets, respectively. $\widetilde{H}_t$ is the candidate hidden state. σ is the sigmoid function after it is activated, it will act as a gate signal after data processing, then reset the gate value on H_{t-1}, it represents how much the amount of information brought by the previous time step can be used, and then use this reset H_{t-1} to perform basic RNN calculations, that is, linear transformation with z_t splicing, and then performing nonlinear activation through the *tanh* function to obtain a new hidden state $\widetilde{H}_t$, and finally, the gate value of the updated gate will act on the new $\widetilde{H}_t$, and the '1-' gate value will act on H_{t-1}, the results of the two are added to obtain the final hidden layer state output H_t. This process means that the update gate can retain the previous results. Above formula $(1 - P_t) \odot H_{t-1}$ represents the selective "forgetting" of the original hidden state, $P_t \odot \widetilde{H}_t$ represents the candidate's hidden state of the current node do selective "remembering". When P_t tends to 1, it means that long-term dependencies always exist, and when P_t tends to 0, which means some unimportant information in the hidden information forgot. After the feature vector passes through the GRU, it can mine the dependencies existing in the sequence while suppressing the gradient disappearance and explosion, and further provid a reliable information guarantee for the anomaly detection stage.

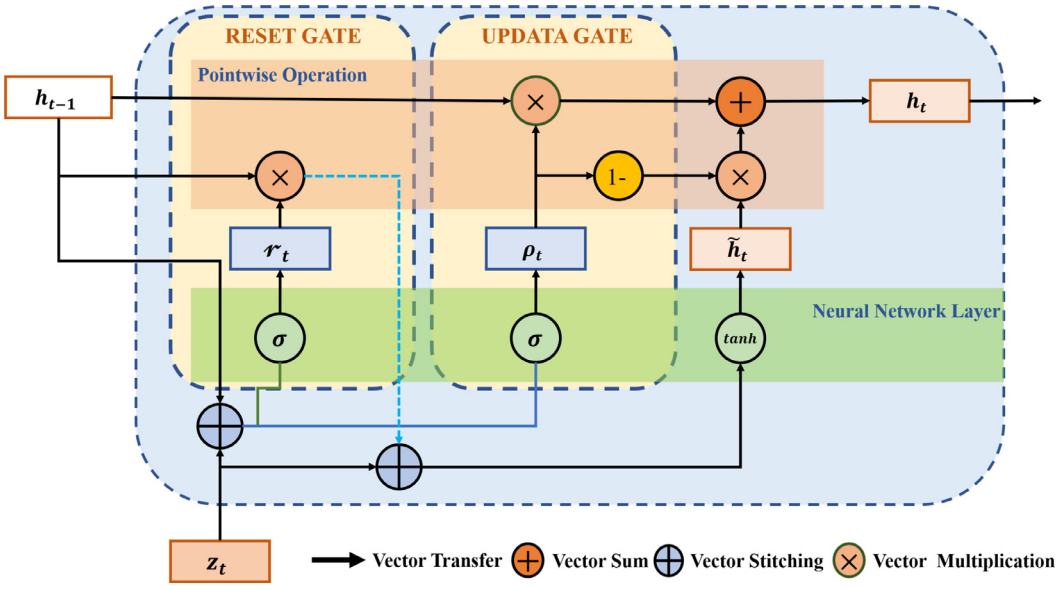


Fig. 2. The internal structure of the GRU model.

Output Layer

From the feature extractor, all N node representations namely $\left\{z_1^{'(t)}, z_2^{'(t)}, \ldots, z_N^{'(t)}\right\}$, are obtianed. Then we multiply it element-wise with the corresponding sensor embedding vector v_i, and use the results of all nodes as the input of a stacked fully-connected layer of output dimension N, to predict the sensor vector behavior value $s_i^{(t)}$ at time step t.

3.4. Abnormal Score

How can we further detect anomalies from the data obtained by the output layer? Usually, when the sensor is abnormal, the relevant behavior data will have apparent deviations compared with the normal behavior data. Therefore, we use this feature to calculate individual anomaly scores for each sensor, which are integral to the subsequent interpretation and detection of anomalies. Then we aggregate these anomaly scores into a single score for each time scale to help users in the detection process. The error value Err is calculated, by comparing the predicted behavior data at time t with the observed behavior data at time t. The value is normalized to avoid the error value being too different. What is more, for the interpretability of anomaly detection, we perform fuzzy positioning of the sensor, calculate the overall anomaly at time t, and use the max function to aggregate the behavioral data of the sensor to help users analyze and determine the general range of abnormal sensors. The formulas involved in the above process are as follows:

$$Err_i(t) = \left|\hat{s}_i^{(t)} - s_i^{(t)}\right| \tag{11}$$

$$q_i(t) = \frac{Err_i(t) - \hat{\mu}_i}{\hat{\sigma}_i} \tag{12}$$

$$Q(t) = \max_i \; q_i(t) \tag{13}$$

where $\hat{\mu}_i$ and $\hat{\sigma}_i$ are the median and interquartile range of the error value $Err_i(t)$ across the time scale, respectively, They are more sensitive to anomalies than mean and variance.

Usually, the sudden change of the value will cause the sensor data to be unpredictable. This phenomenon will likely destroy the data trend and affect the subsequent abnormal judgment. To avoid the situation that the sensor's behavior and value abrupt change can cause abnormal judgment, we use a Simple Moving Average(SMA) to generate a smooth score $Q_s(t)$, and calculate the unweighted value of n values. Arithmetic average, then $Q_s(t)$ is compared with the set decision threshold μ(in the experiment, the threshold μ is set to the maximum value of $Q_s(t)$ on the validation data), If $Q_s(t)$ exceeds μ, the timescale t marks as abnormal; otherwise, it is marked as normal. The formulas involved are as follows:

$$SMA_t = \sum_{i=T-1}^{0} q_{t-i} \tag{14}$$

$$\mu = Max(Q_s(t)) \tag{15}$$

$$y_i(t) = \begin{cases} 1, Q_s(t) > |\mu| \\ 0, Q_s(t) < |\mu| \end{cases} \tag{16}$$

4. Experiment

In this section, we will conduct empirical evaluations to demonstrate the effectiveness of the proposed framework. Specifically, we aim to answer the following research two questions:

RQ1: How effective is the proposed GRN for anomaly detection on correlated multivariate time-series datasets?

RQ2: Can the proposed GRN, in terms of interpretability for anomalies, locate anomalies and help users identify affected sensors, and how can anomalies deviate from expected behavior?

We discuss our results while answering the above research questions as follows.

4.1. Scenario and Attack

- **Scenario**

  The SWaT (Mathur and Tippenhauer, 2016) and WADI (Ahmed et al., 2017) are water treatment test platforms for research in the field of safety. The SWaT water treatment testbed is divided into six sub-processes: Raw Water Storage, Pretreatment, Ultrafiltration, Ultraviolet, Reverse Osmosis, and Disposal. The WADI is a natural extension of the SWaT and includes two elevated tanks, six consumer tanks, two raw water

Table 1
Dataset Statistics.

Dataset	Features	Train	Test	Dimensions	Anomalies
SWaT	51	496800	449919	51(1)	11.98%
WADI	127	1048571	172801	123(1)	5.99%

tanks and one return tank. It is also equipped with a chemical metering system, booster pumps and valves, instrumentation and analyzers. The WADI will absorb a portion of SWaT's reverse osmosis permeate and raw water, resulting in a complete and realistic water treatment, storage and distribution network.

- **Attack**
  The combination of these two testbeds allows researchers to witness the cascading effect of a network attack on a testbed. The relevant attacks covered in this paper are as follows.
  Attack1(SWaT): The level transmitter *HMI_LIT*_101_*PV* is under invisible attack, causing the original water tank level to increase and generate an overflow.
  Attack2(WADI): The Flow Indicator Transmitter 1_*FIT*_001_*PV* is attacked by a false reading. After the PLC gets a false reading, it increases the amount of chemicals in the water in order to keep the water level of the original tank unchanged. At this time, the conductivity measured by Analyser Indicator Transmitter 1_*AIT*_001_*PV* increases. At this time, the conductivity measured by Analyser Indicator Transmitter 1_*AIT*_001_*PV* increases.
  Attack3(WADI): Malicious opening of the regulating control valve 2_*MCV*_007_*CO* creates a leak before the water reaches the consumer tank and the flow rate through pump 2_*P*_003_*SPEED* is reduced.

The actual attack scenario for the operator simulation of the water treatment plant consists of attacks on the PLC and network, as well as simulated physical attacks, such as water leakage and malicious chemical injection. The attacks covered in the explanation phase of this paper are mainly physical attacks.

4.2. Dataset

This experiment will consider two sensor datasets based on water treatment physics testbed systems: SWaT and WADI. We take as input 5 time steps where each sensor produces data. We employ the Adam optimizer for our model training, with learning rates set to 0.1 on SWAT and WADI, respectively, and the step size in both experiments is set to 15 steps. In addition, the *Topk* in our experiment is set to 20. Table 1 summarizes the statistics for each dataset. Details are as follows:

- **SWaT Dataset**
  The SWaT, proposed by iTrust, the Cyber Security Centre of Singapore Polytechnic University, simulates a data set generated by an industrial water treatment plant. SWaT runs in normal mode for seven days and attacks for four days. Finally, physical characteristics related to the water treatment plant and water treatment process generated. The data set of sensor state is used in this experiment together with network traffic and sensor state.
- **WADI Dataset**
  The WADI dataset, curated by the Singapore Public Utilities Commission, comes from natural water treatment, storage, and distribution network. Its data includes sensor measurements from 127 sensors in regular operation for two weeks, used as training data for the model. Continuous process for 16 days: 14 normal operations, two days for abnormal attack scenarios at different time intervals, 104,847 training, 17,270 test data points, and the abnormal rate is 5.99%.

4.3. Comparison Methods

We compare GRN with nine state-of-the-art models for multidimensional time series anomaly detection, including MERLIN (Nakamura et al., 2020), LSTMNDT (Hundman et al., 2018), DAGMM (Zong et al., 2018), OmniAnomaly (Su et al., 2019), MAD-GAN (Li et al., 2019), USAD (Audibert et al., 2020), MTAD-GAT (Zhao et al., 2020), CAE-M (Zhang et al., 2021), GDN (Deng and Hooi, 2021). The details of the baseline methods for these comparisons have been described in Related Work section.

5. Results and Discussion

5.1. Effectiveness Results (RQ1)

In experiments, we evaluate the performance of the proposed framework GRN by comparing it to the included baseline methods. Table 2 shows the anomaly detection results of different models measured by Precision, Recall, and F1 metrics on SWaT and WADI datasets. In our experiments, we set the number of epochs of GRN to 50 and 100, denoted as GRN-50 and GRN-100, respectively. It turns out that in both SWaT and WADI datasets, we get the observations:

- **Precision**
  The accuracy of GRN-50 outperforms all baseline models (except USAD on the SWaT dataset). This situation is because USAD uses an autoencoder with two decoders and an adversarial training framework, which is especially suitable for the stable SWaT dataset, bringing high detection accuracy while ensuring good performance. GRN-100 outperforms all other state-of-the-art methods in accuracy on both datasets. Meanwhile, on SWaT, our model achieves apparent advantages on the AUC metric.
- **AUC and F1 score**
  We can observe that as the number of epochs increases, the F1 score of GRN on both datasets improve to a certain extent, where GRN-100's F1 score on WADI is better than that of the baseline model. In terms of AUC results, GRN's AUC on the SWaT dataset leads all other baseline models, while on the WADI dataset with higher instability and dimension, the AUC slightly lags.
- **Recall**
  By comparing the experimental results of GRN on the SWaT dataset with 50 epochs and 100 epochs, we observe no significant fluctuations in both. However, on the WADI dataset with an enormous amount of data and higher dimensions, the detection results of GRN-50 and GRN-100 fluctuate considerably, and there is a certain instability. In addition, the recall value of GRN-100 is significantly improved compared to GRN-50.

5.2. Model Interpretability (Q2)

- **Interpretability of Embedding Vectors for Sensors**
  We plot the sensor vector structure clusters learned on the SWaT and WADI datasets as t-SNE diagrams for the interpretability of embedding vectors, as shown in Figure 3(a)(b), respectively. We shade the vector nodes for these sensors. The vector node colours of the same class of sensors are the same, and the colours of different classes are different. In addition, through Figure 3(a)(b), we observe the local clustering of these sensors projected in 2D space, as indicated by the dotted lines. These sensors perform the functions of the relevant indicators in the tank in both SWaT and WADI water distribution networks, which explains the strong similarity in the existence of these local cluster sensors. This graph can help users analyze

Table 2
Performance comparison of GRN method and baseline method on the whole dataset. P: precision, R: recall, AUC: area under the ROC curve, F1: F1 score, complete training data. Best precision is highlighted in bold.

Method	SWaT				WADI			
	P	R	AUC	F1	P	R	AUC	F1
MERLIN	0.656	0.2547	0.6175	0.3669	0.0636	0.7669	0.5912	0.1174
LSTM-NDT	0.7778	0.5109	0.714	0.6167	0.0138	0.7823	0.6721	0.0271
DAGMM	0.9933	0.6879	0.8436	0.8128	0.076	0.9981	0.8563	0.1412
OmniAnomaly	0.9782	0.6957	0.8467	0.8131	0.3158	0.6541	0.8198	0.426
MAD-GAN	0.9593	0.6957	0.8463	0.8065	0.2233	0.9124	0.8026	0.3588
USAD	0.9977	0.6879	0.846	0.8143	0.1873	0.8296	0.8723	0.3056
MTAD-GAT	0.9718	0.6957	0.8464	0.8109	0.2818	0.8012	0.8821	0.4169
CAE-M	0.9697	0.6957	0.8464	0.8101	0.2782	0.7918	0.8728	0.4117
GDN	0.9591	0.6957	0.8462	0.8101	0.2912	0.7931	0.8777	0.426
GRN-50	0.9972	0.5921	**0.8781**	0.7389	**0.9651**	0.2497	0.7832	0.3981
GRN-100	**0.9986**	0.5909	**0.8845**	0.7496	**0.3584**	0.7398	0.78212	**0.4828**

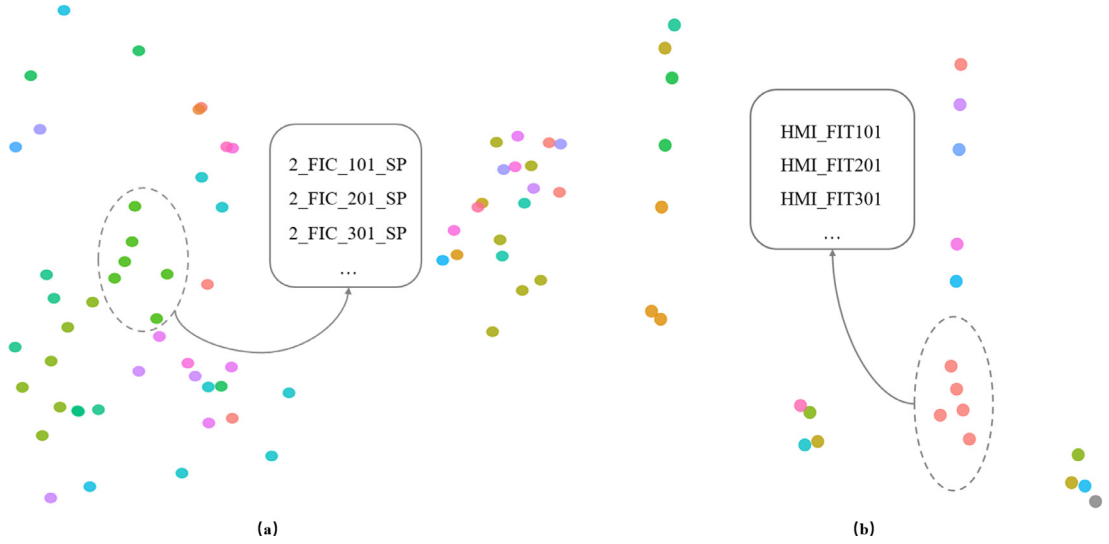


Fig. 3. t-SNE plots of sensor embeddings of our trained model on SWaT and WADI datasets (dimensionality reduction by PCA, normalized), respectively. Node colors indicate classes. The stippled areas represent the local clusters of *HMI_FITX*01 sensors and 2*_FIC_X*01*_SP* sensors, respectively. These sensors measure similar metrics on SWaT and WADI.

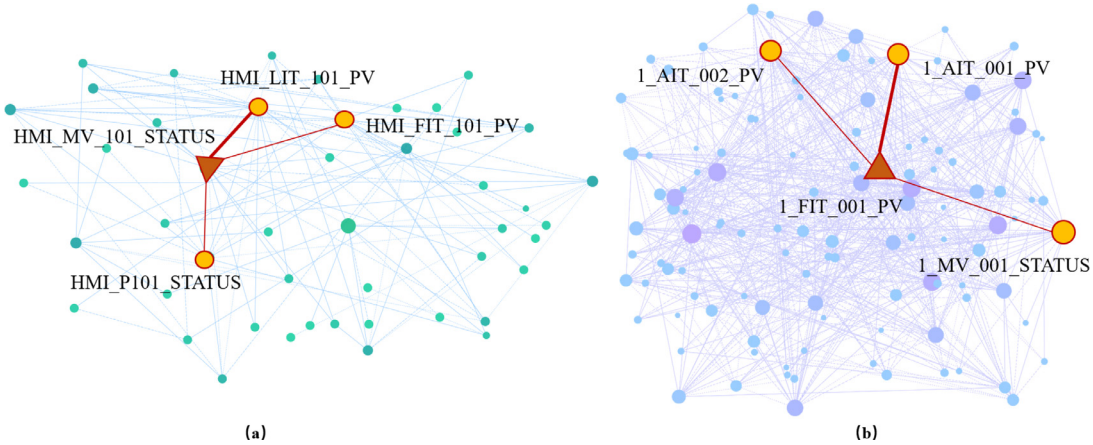


Fig. 4. Undirected graphs of weighted networks generated by sensor-in-graph structure learning in SWaT (left) and WADI (right). Yellow circles indicate nodes with edge weights greater than 0.1 from the central node.

the dependencies sensors, and it is indispensable for anomaly interpretability.

- **Interpretability of Attacks**
  The first anomaly case is about SWaT, which is an invisible attack on HMI LIT 101 PV, causing the original tank to be in the overflow state. As shown in Figure 4(a), the red triangle is the sensor with the highest anomaly score from GRN for deviating sensors, indicating that *HMI_MV*_001*_PV* may be the attacked sensor or closely related to the attacked sensor. *HMI_LIT*_101*_PV* is highly related to *HMI_MV*_001*_PV*, which controls the sensor that is used by the former to measure and transmit the raw water volume, level height of the valve. As

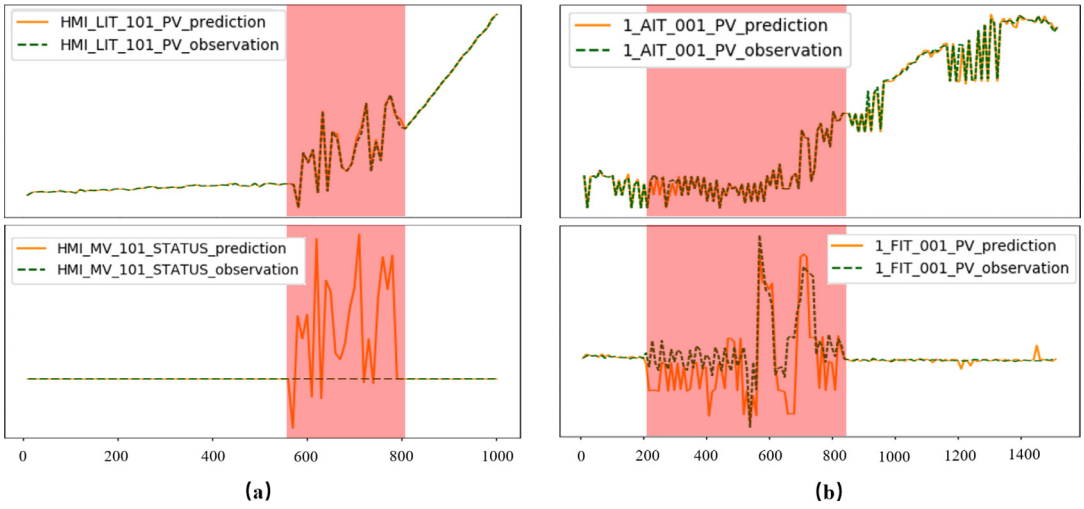


Fig. 5. Comparing predicted and observed data for SWaT (left) and WADI (right), respectively, helps to explain the anomalies. The attack period is shaded in red.

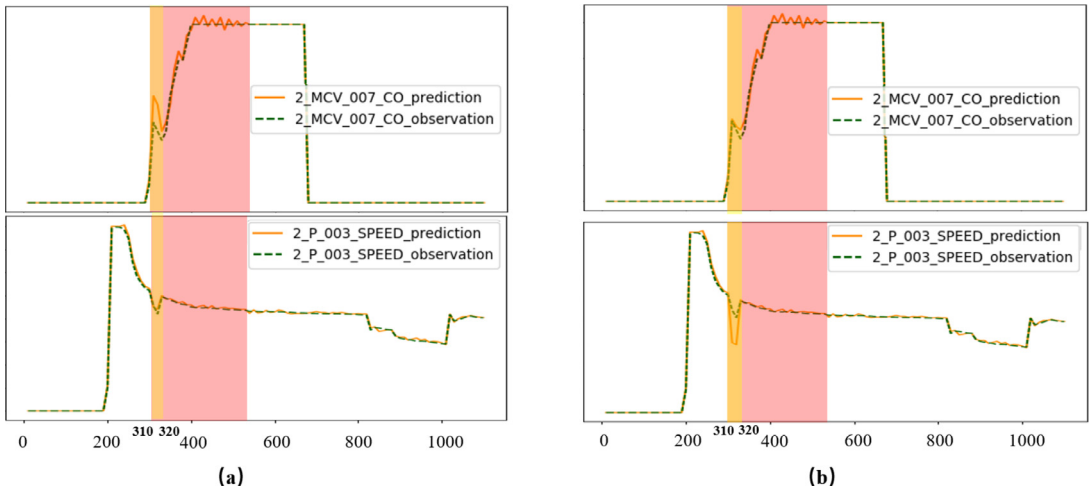


Fig. 6. Comparison of expected and observed data for the sensors of interest in the WADI dataset for GRN (left) and GDN (right). The attack period is shaded in red. The orange shading shows the contrast difference during the attack period, corresponding to the layout of the guide diagram in Figure 7.

*HMI_LIT*_101_*PV* is under invisible attack, the raw water tank gradually rises to the overflow state, when the flow rate measured by *HMI_LIT*_101_*PV* tends to increase in general. The anomaly is further understood by comparing the predicted and observed sensor values in Figure 5(a): for *HMI_MV*_001_*PV*, there is a significant deviation from the actual observed sensor values due to the predicted values of the attack model, which leads to an error that causes GRN to judge it as an anomaly.

The second anomaly case is about WADI, which is a false reading attack on 1_FIT_001_PV, causing the chemical pump to inject high doses of chemicals into the raw water tank to maintain the water level. As shown in Figure 4(b), 1_*FIT*_001_*PV* is identified as the deviating sensor with the highest anomaly score. 1_*AIT*_001_*PV* is highly correlated with 1_*FIT*_001_*PV*. 1_*AIT*_001_*PV* is responsible for measuring the conductivity of the raw water tank, and 1_*FIT*_001_*PV* measures the flow state of the chemical flow in the raw water tank by the latter. The conductivity measured by 1_*AIT*_001_*PV* tends to increase overall, because 1_*FIT*_001_*PV* is attacked by false readings, resulting in an increase in the chemical content in the water. The anomaly is further understood by comparing the predicted and observed sensor values in Figure 5(b): for 1_*FIT*_001_*PV*, both of the model prediction and the actual observed sensor values show more pronounced anomalous fluctuations, causing GRN to judge them as anomalous.

- **Interpretable advantages of GRN compared to GDN**
  We analyze and compare an abnormal area on the WADI. The anomaly is actually caused by 2_*MCV*_007_*CO* being turned on maliciously, which may lead to mainline leakage. Figure 6(a)(b) shows the predicted and actual sensor values of GDN and GRN for this region, respectively. 2_*MCV*_007_*CO* in GDN shows deviation and 2_*P*_003_*SPEED* in GRN shows deviation in the time period 310 to 320. We compare the interpretation of GRN with GDN for the time period 310 to 320, as shown in Figure 7(a)(b). 2_*MCV*_007_*CO* is identified by GDN as the sensor with the highest abnormal score, while 2_*P*_003_*SPEED* is identified by GRN as the sensor with the highest abnormal score. Above all, we conclude that the interpretability ability of GDN is weaker than that of GRN.

5.3. The solution of gradient loss and gradient explosion problem

In Figure 8(a), the overall trend of the AUC of GRN is similar to that of GDN, while the AUC of GRN is less than that of GDN. In addition, Figure 8(b) shows that when dealing with more complex WADI, GDN has the problem of gradient explosion during

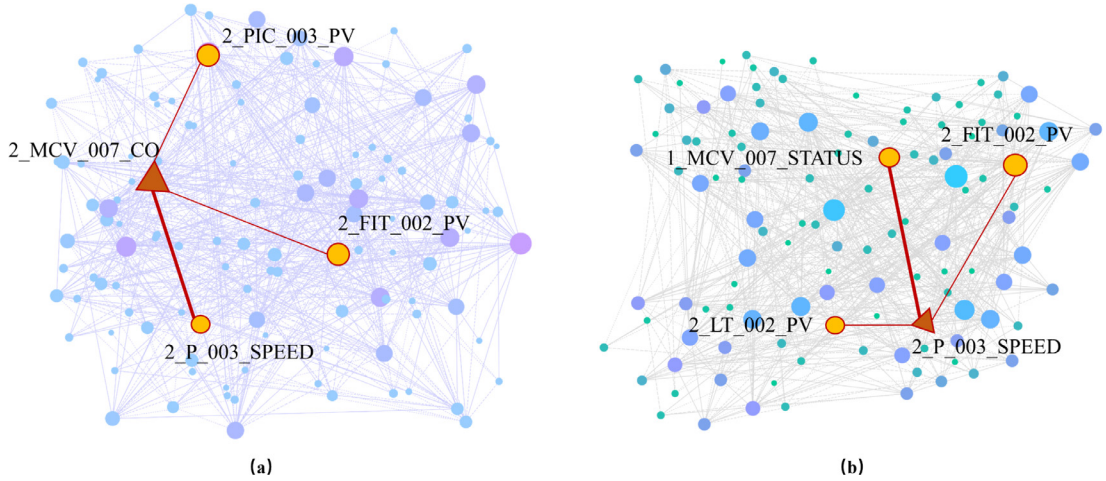


Fig. 7. The learning of the structure within the sensor graph generates an undirected graph of the weighted network, showing the attacks of GRN(left) and GDN (right) in WADI, respectively.

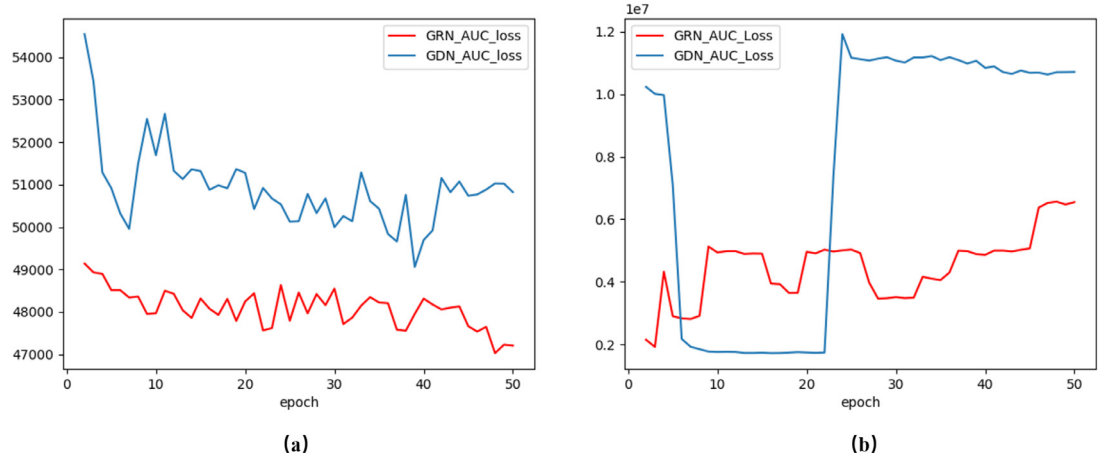


Fig. 8. (a) (b) are the line graphs of the AUC of the GRN and the baseline model GDN under the setting of epoch=50 on the SWaT (left) and WADI (right) datasets, respectively.

the training process and gradient disappearance at the later stage. While GRN effectively avoids the gradient disappearance and explosion problems.

6. Conclusion

In this paper, we make the following two key contributions. First, we propose GRN, an interpretable anomaly detection method for multivariate time series, which can not only automatically learn complex dependencies between variables with graph neural network, but also mine deeper dependencies between variables, combine with attention mechanism and GRU model, making it easier to identify and interprete abnormal behavior. Our results show that GRN has significant detection precision and recall overcoming existing nine state-of-the-art algorithms in the field of multivariate time series anomaly detection. In addition, the model has valid interpretability to help users find the actual root cause of anomalies, and locate anomalous sensors effectively. Compared to GDN, the traditional interpretable multivariate time series anomaly detection algorithm, the experimental results demonstrate that our model has better interpretable ability, and further effectively solves the gradient vanishing and gradient exploding problems, that occur during training.

There are essentially two limitations to this experiment. First, we did not consider the drawbacks of the GRU model. It cannot be calculated in parallel, which limits the performance of GRN. Second, in the process of anomalous attacks on sensors, our model cannot still judge whether the attack has stopped or not. In future work, for the first limitation, we will explore different parallel mechanisms and try additional architectures or model to improve the detection efficiency. Besides, for the second limitation, we will perform efficient anomaly detection in complex scenarios.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Chaofan Tang: Conceptualization, Methodology, Software, Writing – original draft. **Lijuan Xu:** Resources, Project administration. **Bo Yang:** Validation, Investigation. **Yongwei Tang:** Software. **Dawei Zhao:** Supervision, Writing – review & editing.

Data Availability

The data that has been used is confidential.

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